CSC336 WEB TECHNOLOGIES · MIDTERM EVIDENCE

Tesla Vehicles

Mega-navigation, validated checkout, and JavaScript cart

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Registration: SP24-BCS-156

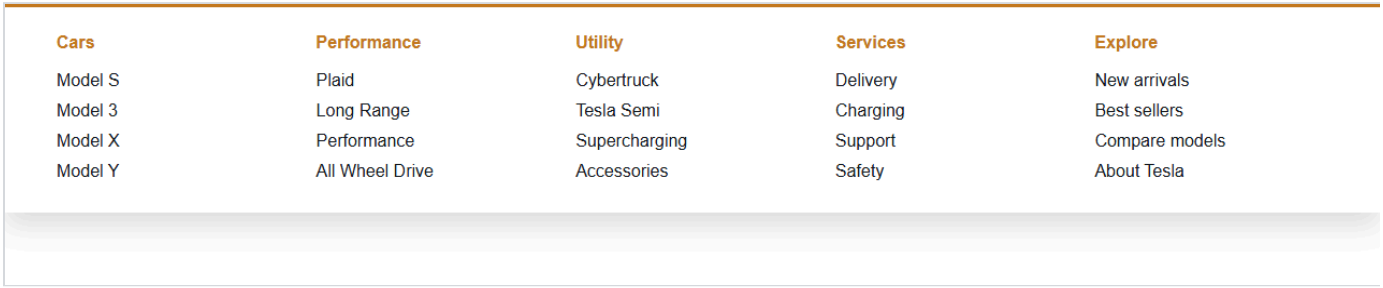
Reference: Tesla — tesla.com

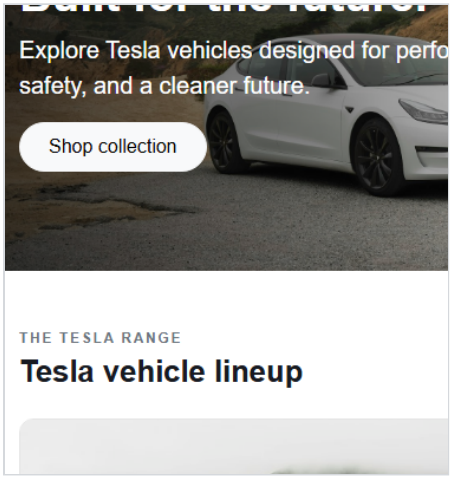
Personalized values: N = 56 · BP = 656px · H = 32° · K = 5

This PDF documents the completed midterm work in the **midterm/** folder.

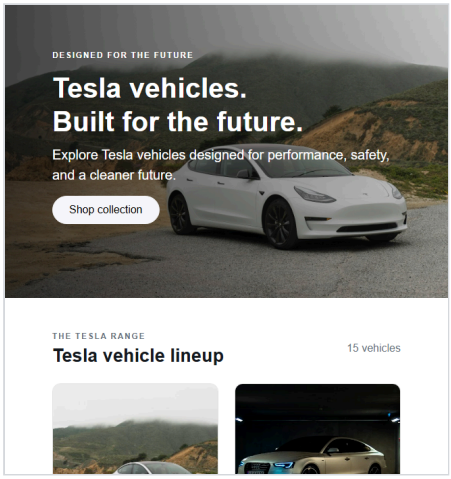
Task 1 · Mega-navigation

The desktop navigation presents five category columns. The tablet and mobile layouts use the same groups without pushing the product content downward; below 656px, the groups are native closed-by-default `<details>` accordions.

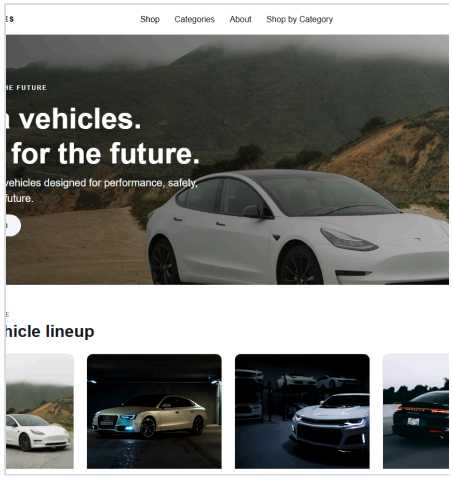




Phone · 375px



Breakpoint · 656px

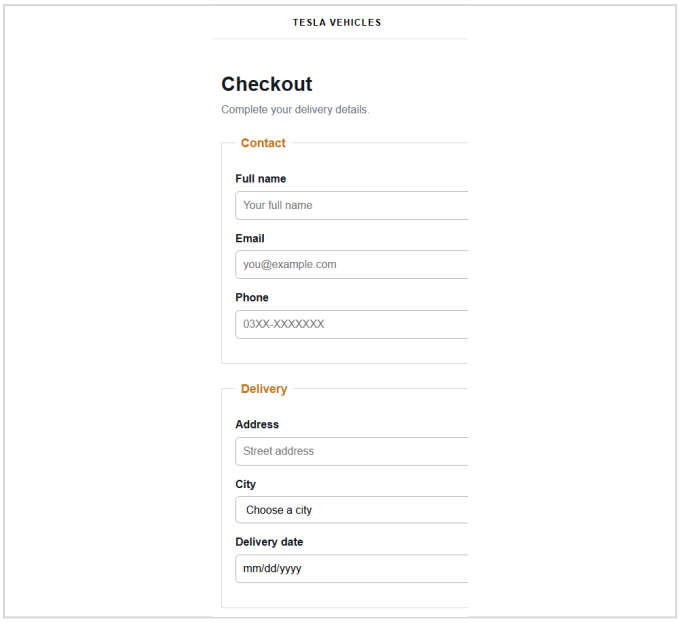


Desktop · 1280px

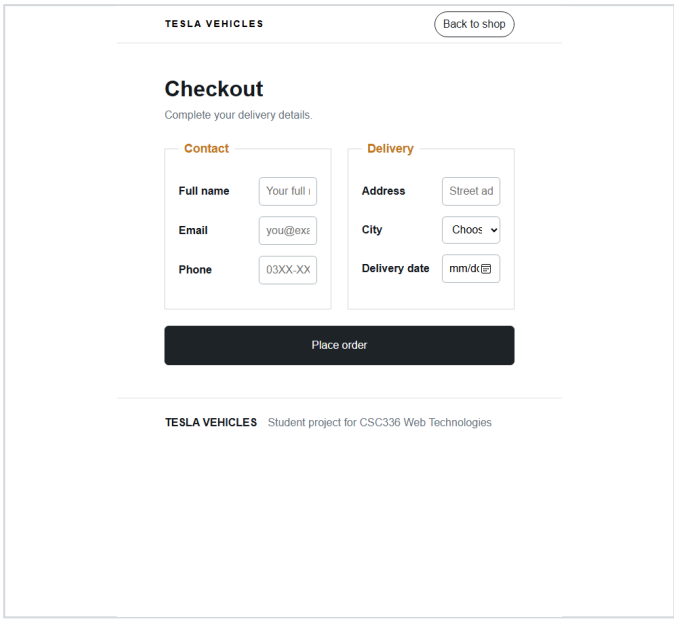
> **Implementation:** CSS Grid creates the five columns; `:hover` and `:focus-within` reveal the desktop panel. The mobile groups use semantic `<details>` and `<summary>`.

Task 2 · HTML5 checkout

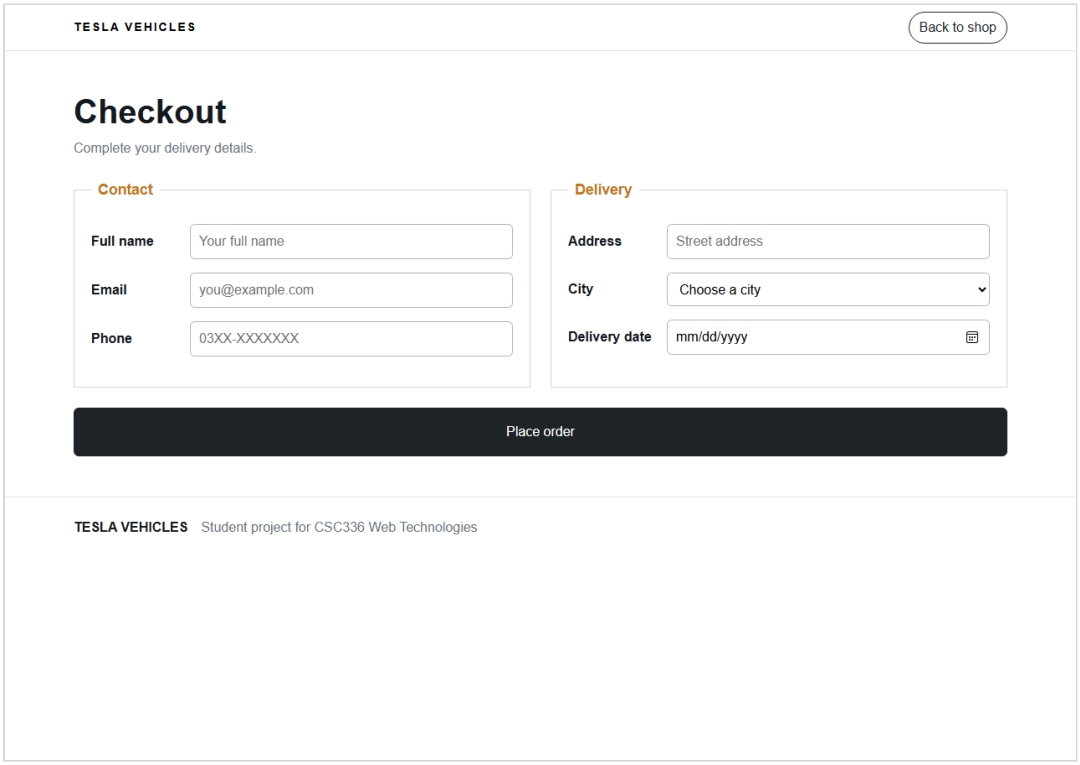
The checkout is implemented without JavaScript. Labels are connected to controls, both fieldsets have legends, and browser validation handles required fields, email type, phone pattern, and the exam-date range.



Phone · stacked fields



Breakpoint · two columns

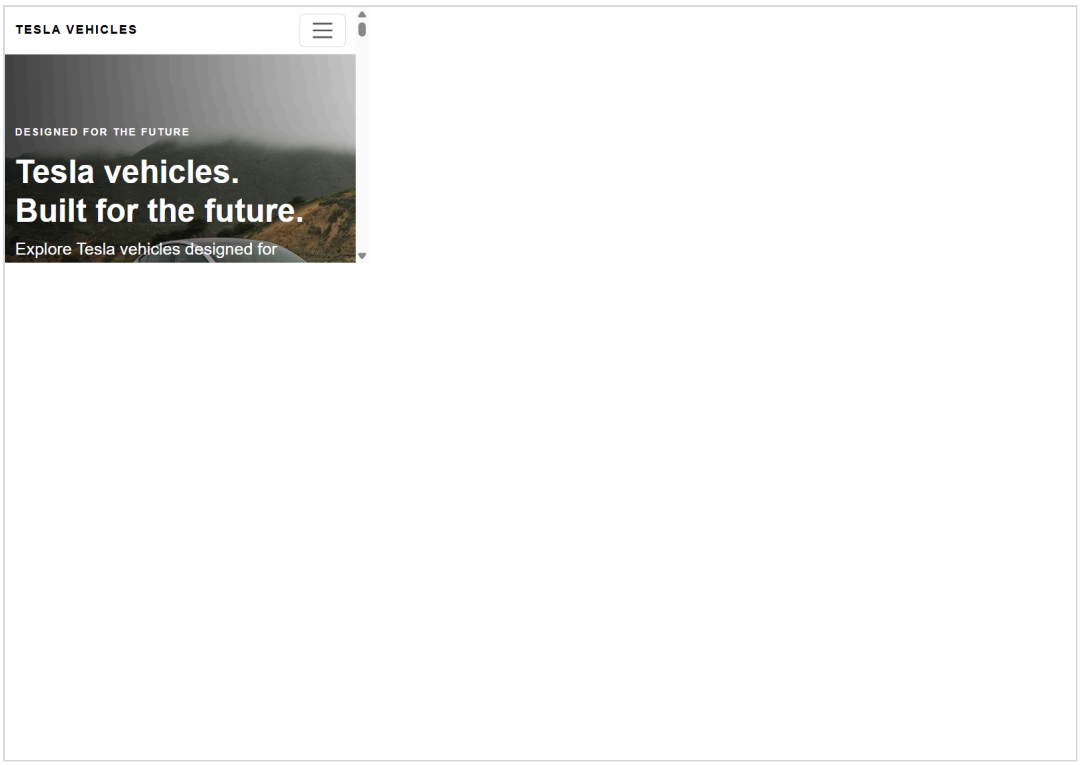


Desktop · aligned labels and inputs

> **Validation rules:** phone format `03XX-XXXXXXX`; delivery date from `2026-10-29` through `2026-12-24`; all six customer fields are required.

Task 3 · JavaScript cart

The cart reads the product name and price from the existing cards, stores items in an array, updates the badge, renders a removable mini-cart, and calculates the displayed total with `reduce()`.



Verified behavior

- Two Model S additions are accepted.
- A third addition is blocked.
- The button changes to “Limit reached”.
- Mini-cart total displays $179,980.00.

Personalized limit

`(N mod 4) + 2 = (56 mod 4) + 2 = 2`

The limit is applied per product, not across the whole cart.

> **Source:** `script.js` uses event listeners, an array for cart state, a remove action, and a `reduce()` total. It is loaded with `defer`.

Submission checklist

Files included

- `index.html`
- `product.html`
- `checkout.html`
- `styles.css`
- `script.js`
- `README.md`
- local Tesla vehicle images
- this completion PDF

Requirements covered

- Five desktop mega-menu columns
- Five mobile accordion groups
- 656px responsive breakpoint
- HTML5-only checkout validation
- CSS Grid checkout layout
- Plain JavaScript cart behavior
- Per-product maximum of two

> **Student explanation:** The implementation uses familiar HTML, CSS, Bootstrap layout classes, and plain JavaScript. Each task is separated into a small, explainable responsibility rather than relying on a complex framework.